xhuang@mae.carleton.ca

Functions of Thermal Barrier Coatings and Their Impact on Gas Turbine Component Life

Xiao Huang, P. Eng., Ph.D.
Associate Professor
Dept. of Mechanical & Aerospace Engineering
Carleton University

© Xiao Huang, Oct. 2008

xhuang@mae.carleton.ca

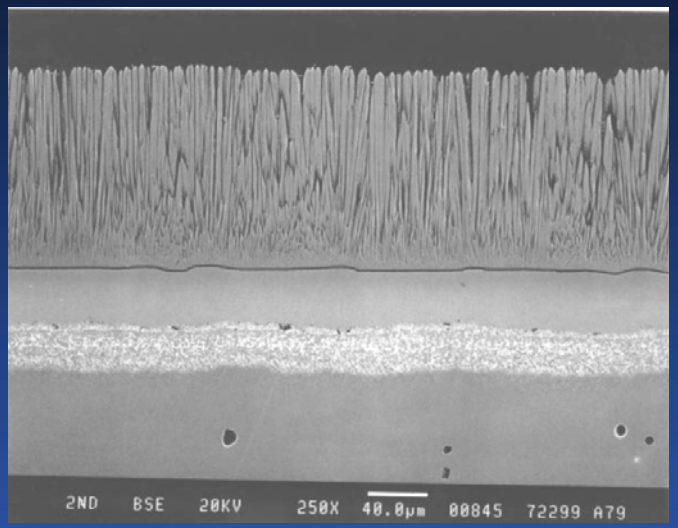
Outline

- Functions and development of thermal barrier coating systems
- Compositions and microstructures of thermal barrier coatings
- Failure mechanisms
- Life of thermal barrier coating system
- Limitations and future directions
- Discussions

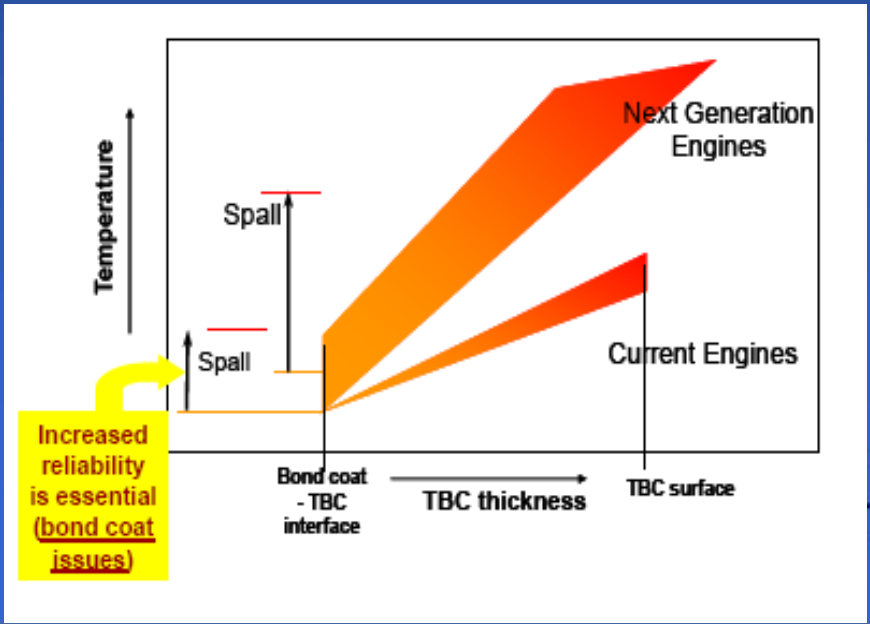
Carleton UNIVERSITY

xhuang@mae.carleton.ca

Thermal Barrier Coating (TBC) in GT



- A coating system providing heat shielding, corrosion, oxidation and erosion resistance
- 4 layer complex, dynamic system

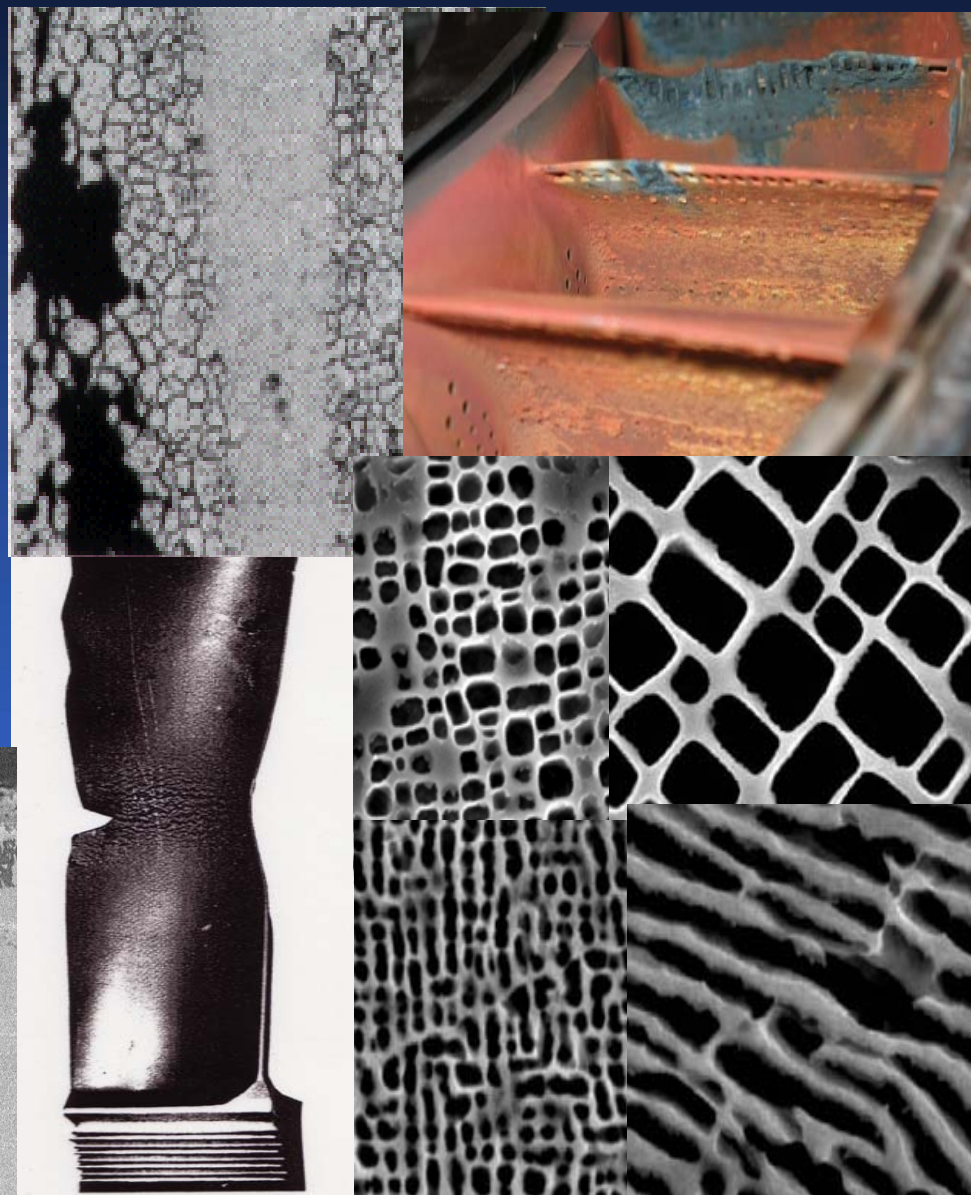


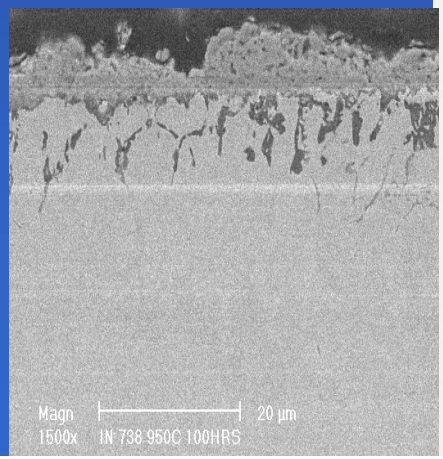
Subramania et al, TurboExpo 2008 GT2008-51532
Clarke and Levi, Annu. Rev. Mater. Res. 33(2003)383–417

xhuang@mae.carleton.ca

Failure Mechanisms of GT Components

- Overheating (burning(√)
- Fatigue (√)
- Creep (>0.4Tm) (√)
- Oxidation (√)
- Hot Corrosion from salt deposit (√)
- Erosion (√)
- Base metal microstructure deterioration (aging, rafting, TCP phase formation) (√)
- Wear

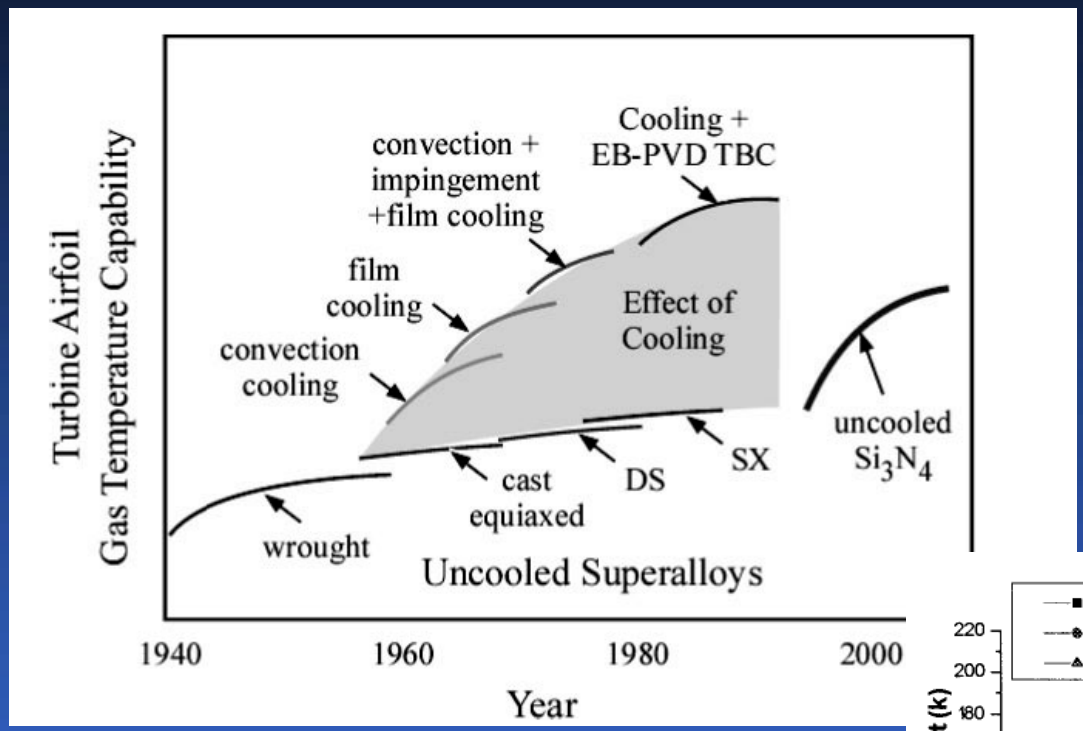




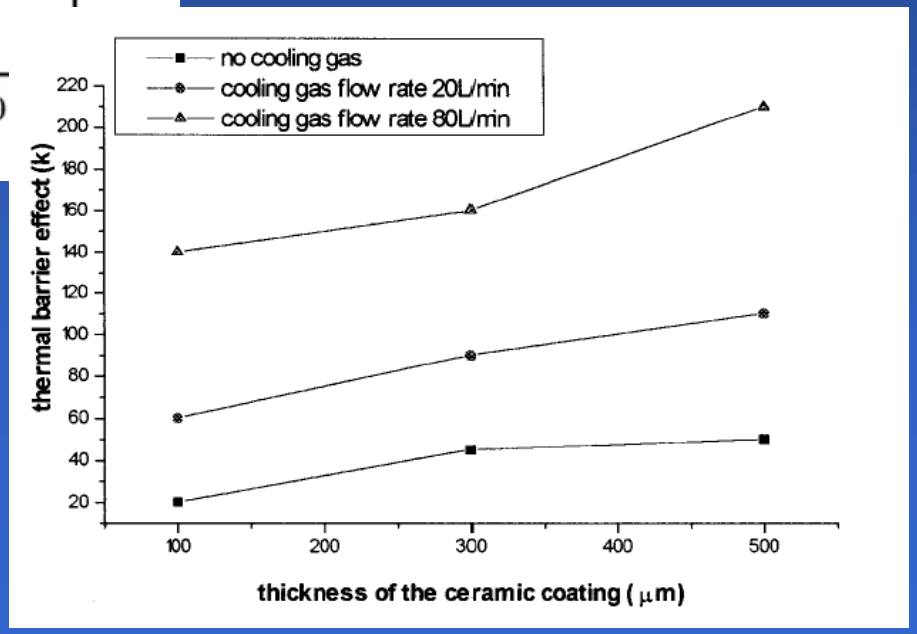
TurboExpo 2006, GT2006-90137
TurboExpo 2006, GT2006-90075
Nagy, et al, Liburdi Turbine Services
X. Huang, AERO 3700, Course Notes, Carleton University

xhuang@mae.carleton.ca

Historical Development in GT Materials



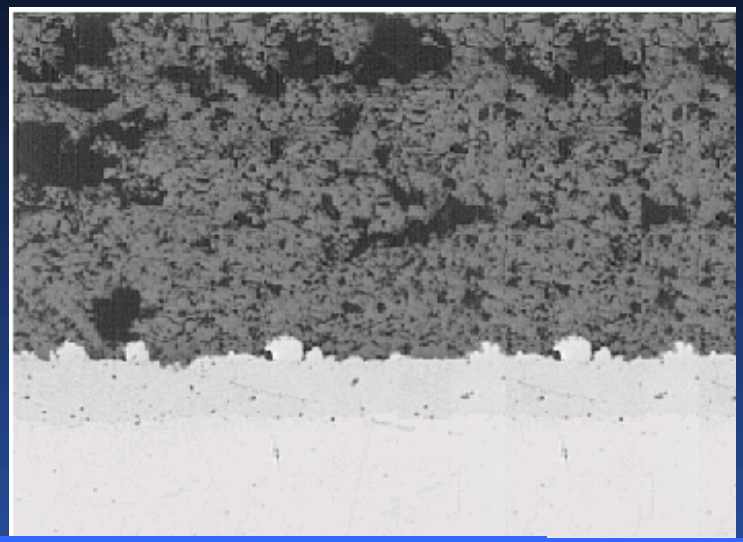
50ºC reduction on metal surface = 50 yr superalloy development

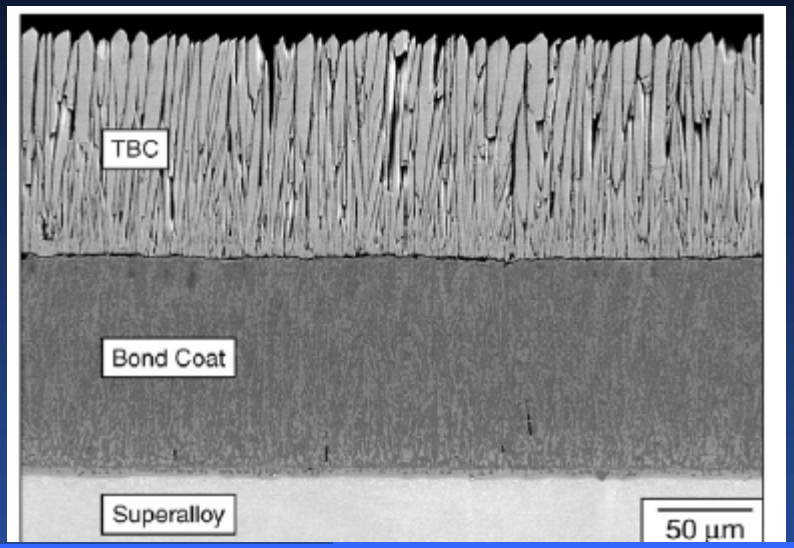


Clarke and Levi, Annu. Rev. Mater. Res. 33(2003)383–417
H. Xu et al, Thin Solid Films, 334(1998)98-102.

xhuang@mae.carleton.ca

Historical Development in TBC





Gen I -1970s
Plasmas sprayed (PS) MCrAlY and 7YSZ on combustion can & afterburner
Black fracture surface
Thick porous laminar TBC (~1 mm)
Up to 500 K reduction when combined with cooling

Gen II - 1980s
Low-pressure plasma-sprayed (LPPS) MCrAlY and PS 7YSZ on high pressure turbine nozzles and transition ducts
White failure mode (cracking in TBC)

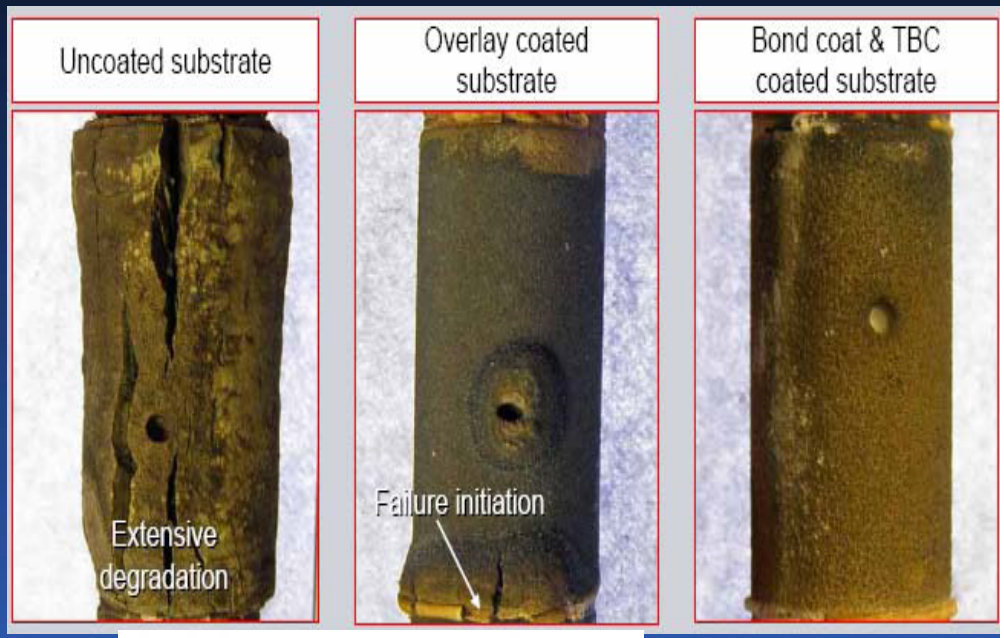
Gen III - Late1980s
LPPS NiCoCrAlY or Pt modified aluminide and EB-PVD columnar 7YSZ on rotational components
Typically 250 µm thick
Failure in TGO or interfaces

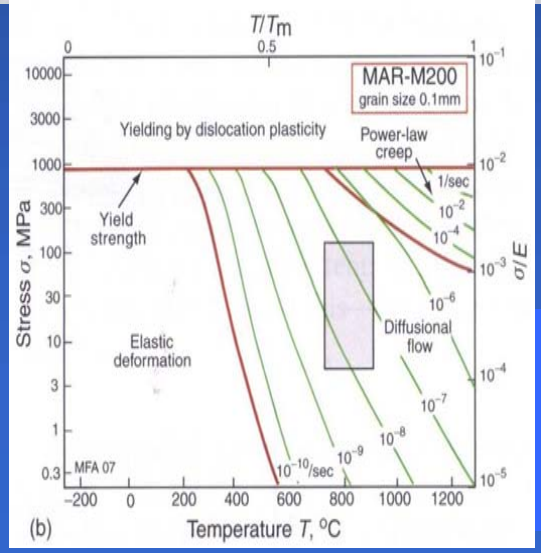
Maricocchi et al., J. of Thermal Spray Tech. vol. 6(2)(1997)193-198 (GE Aviation)
Bose et al., J. of Thermal Spray Tech. vol. 6(2)(1997) 99-104 (Pratt & Whitney)

xhuang@mae.carleton.ca

Functions of Top and Bond Coats

- 7YSZ top coat (TC)
  - Increase creep life of substrate by reducing steady state metal surface temperature
  - Reduce thermal stresses during transient stage (heat up and cool down)
  - Environmental protection
  - Increase bond coat creep life (reduced rumpling of bond coat at >DBTT)
  - Reduce cooling air requirement
  - Enable high TIT (efficiency and emission)
- MCrAlY or aluminide bond coat (BC)
  - Oxidation and corrosion resistance
  - Bonding of top coat to substrate
- Coating life determines inspection interval





20ºC=100% increase in life

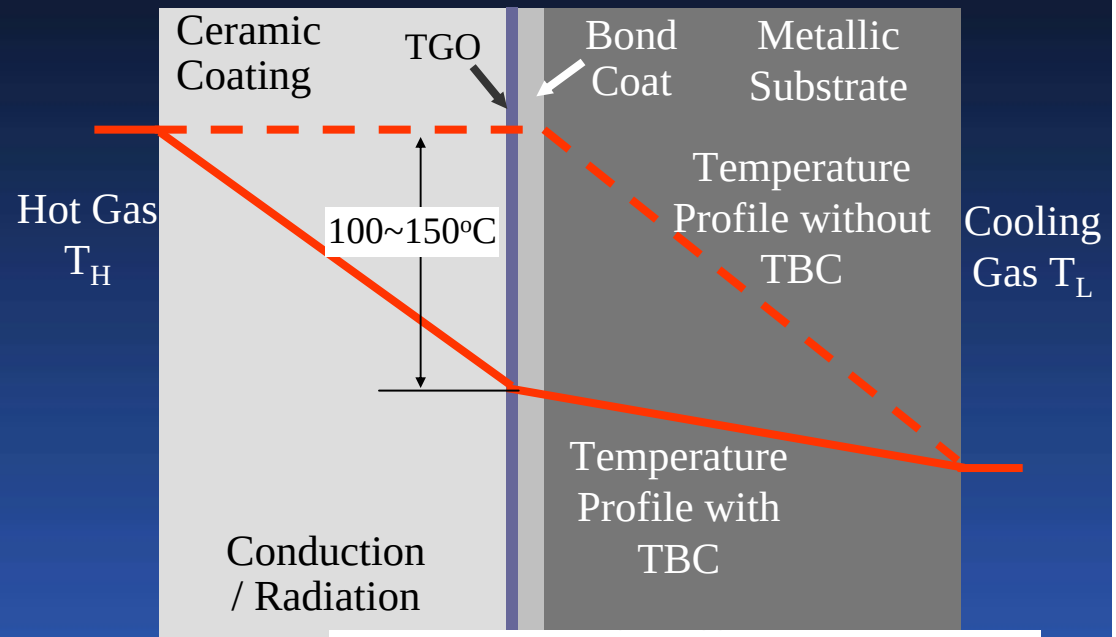
Subramania et al, Turbo Expo 2008, GT2008-51532
Ashby et al, Materials-engineering, science, processing and design, 1st ed. Elsevier, 2007

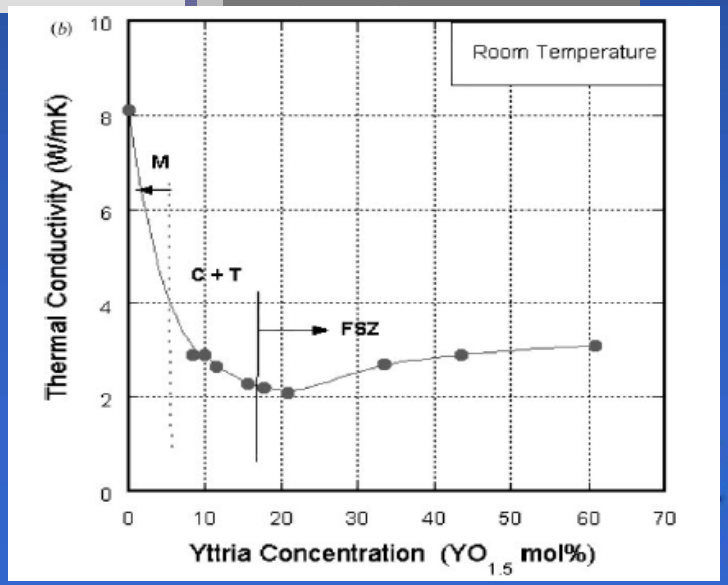
xhuang@mae.carleton.ca

Requirements of Top Coating

Requirement for TBCs

- Thermal insulation
- High melting points
- Microstructure stability
- Fracture toughness
- Matching thermal expansion coefficient
- Low elastic modulus





7wt% Y_2O_3 partially stabilized ZrO_2 (7YSZ)

(porous or vertically cracked structure to accommodate strains)

Clarke et al, Annu. Rev. Mater. Res. 33(2003)383–417

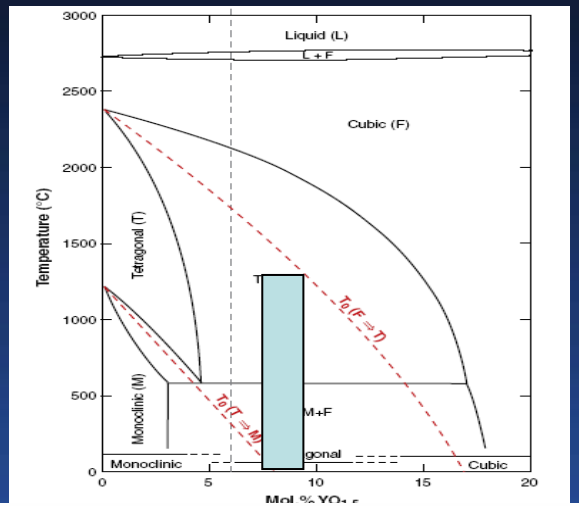
D. M. Wang et al. Canadian Aeronautics and Space Journal, Vol.50 (2)(2004)107-114.

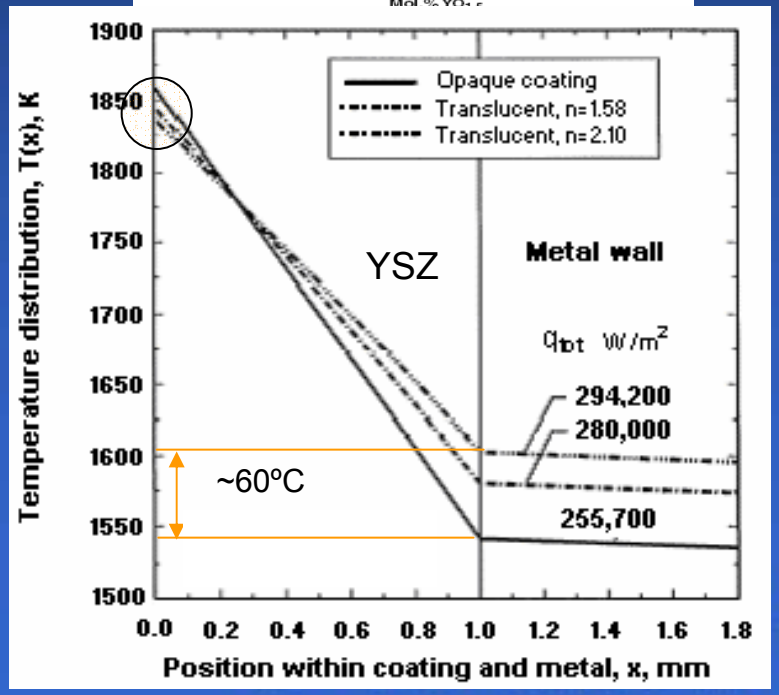
xhuang@mae.carleton.ca

Deficiencies of 7YSZ

- Fast cooling from APS or EB-PVD →t'
- Transparent to thermal radiation
- Increased thermal conductivity and elastic modulus upon thermal exposure (sintering) at 1200ºC
- Changing thermal stresses
- Decomposition of t' phase at 1200ºC leading to cracking

  t' → t + c → m + c
- High oxygen diffusivity - oxidation of bond coat



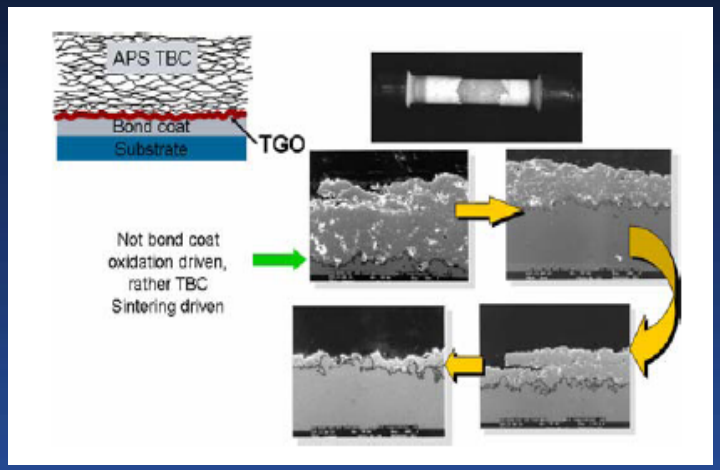


Chevalier, Annu. Rev. Mater. Res. 37(2007)1-32
Siegel et al, Materials Science and Engineering, A245(1998)150-159.
D. M. Wang et al, Canadian Aeronautics and Space Journal, Vol.50 (2) (2004)107-114.

xhuang@mae.carleton.ca

Failure Mechanisms for TBC Top Coat

- TBC system prone to failure under thermal cycling
- TBC failure – immediate component failure from overheating for modern engine
- Typical failure modes:
  - FOD or erosion
  - Top coating cracking (excessive temperature – phase transformation and sintering related cracking)
  - CMAS damage
  - Hot corrosion of top coat (leaching out yttria and phase transformation)
  - Top coat spalling (TGO formation and cracking at the interface- further discussed in BC failure)
- Coating process, operating temperature, time and cycle dependent
- Mostly experiment based life predication approach



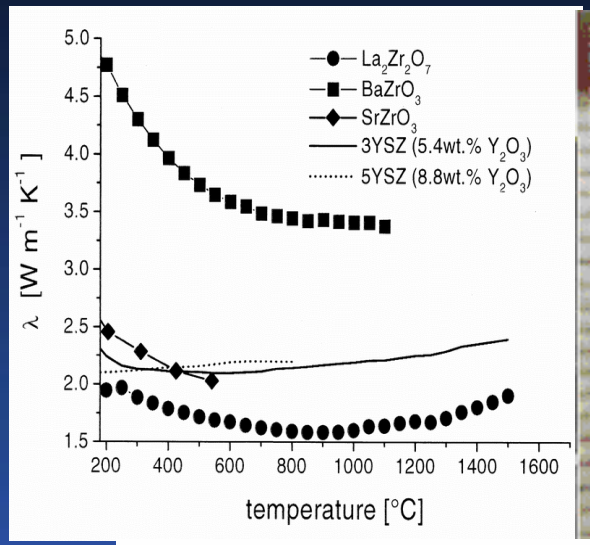
At higher operating temperature, top coat layer-by-layer de-lamination due to sintering effect

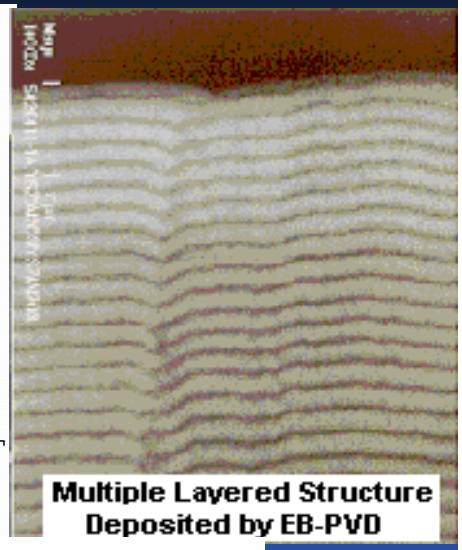
Subramania et al, Turbo Expo 2008, GT2008-51532
S. Kramer et al, J. Am. Ceram. Soc., 91(2)(2008), 576-583
Clarke et al, Annu. Rev. Mater. Res, 33(2003)383-417

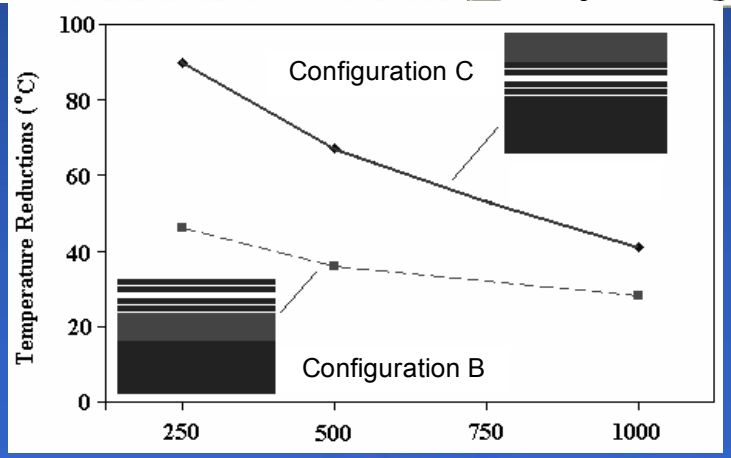
xhuang@mae.carleton.ca

New TBC Top Coat

- Requirements
  - Lower thermal conductivity
  - Stable structure
  - Improved fracture toughness
- Modification to existing 7YSZ
- New structures (layers)
- New coating compositions
- Stable at higher temperature
- Promising directions
  - EB-PVD multiple layered structure
  - Multiple dopants
  - Rare earth zirconate







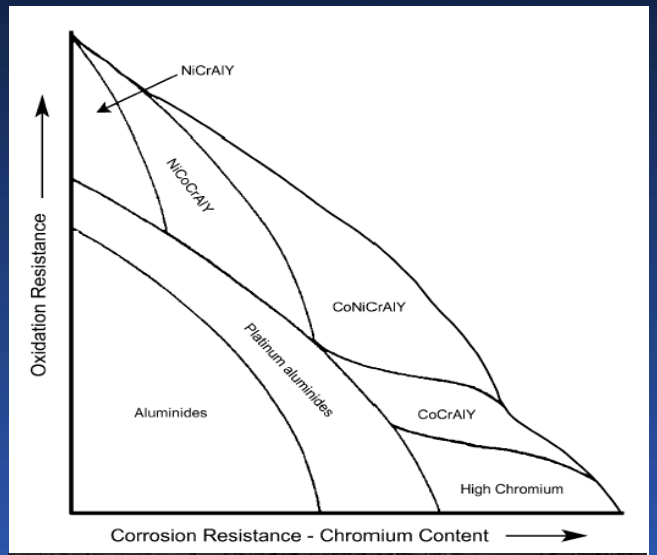
Maloney MJ. 2000. U.S. Patent No. 6,117,560
Vassen et al. *J. Am. Ceram. Soc.* 83(2000) 2023–28
X. Huang, J. Eng. Gas Turbines and Power, 2009, Vol. 131
J. singh et al, Penn state University

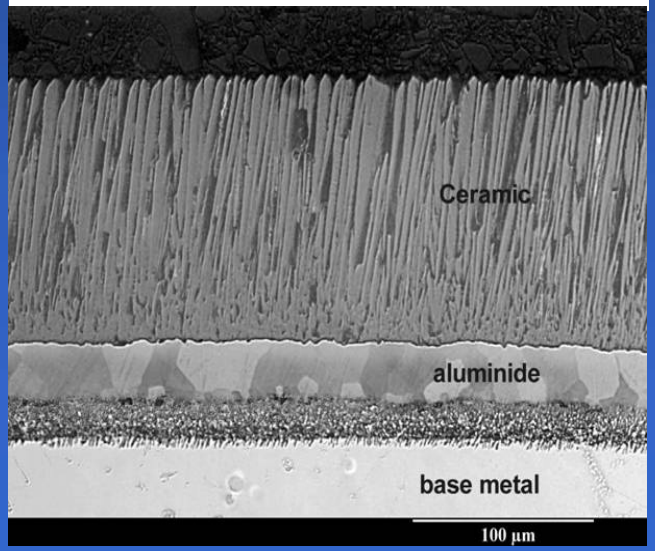
Carleton UNIVERSITY

xhuang@mae.carleton.ca

Bond Coat

- As bond coat or stand alone
- Aluminide (β-NiAl, (Ni,Pt)Al, $PtAl_2$+(Ni,Pt)Al)
  - High DBTT (600-900ºC)
  - CVD, pack cementation, slurry
  - Pt plating+ VPA
  - DVD NiAlPt alloy
  - Diffusion HT
- MCrAlY overlay
  - β-Ni(Fe,Co)Al + γ
  - Lower DBTT (200-700ºC)
  - 2-4 X higher cost
  - Coated component has service life 2-5 times of bare component
  - NiCrAlY-oxidation resistant
  - CoCrAlY-hot corrosion resistant
- Compositions
  - Al: high temperature oxidation resistance
  - Cr: hot corrosion to sulphur containing salts
  - Y: oxide film adhesion
  - Pt: slow diffusion and improve oxidation resistance
  - Si: type II hot corrosion resistant

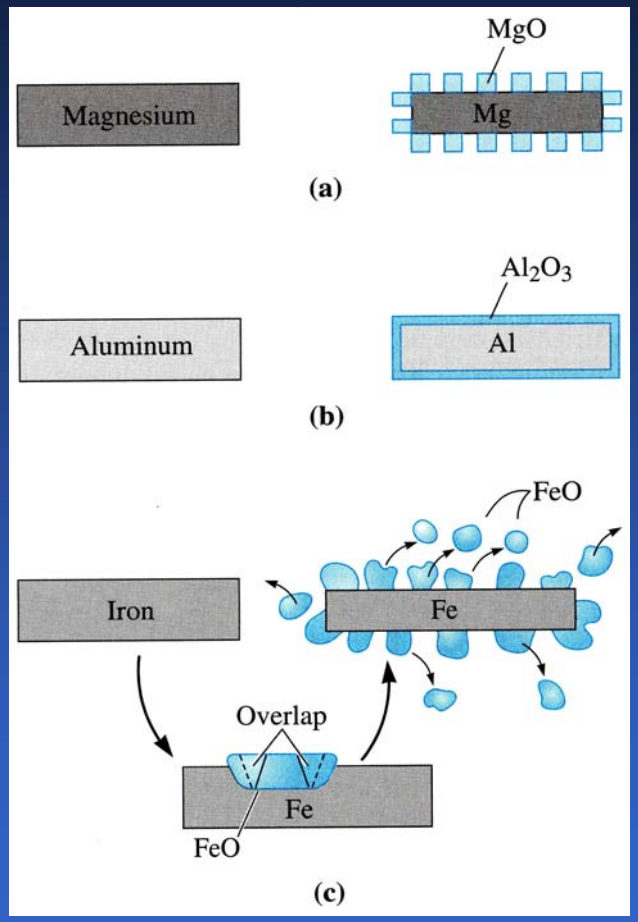


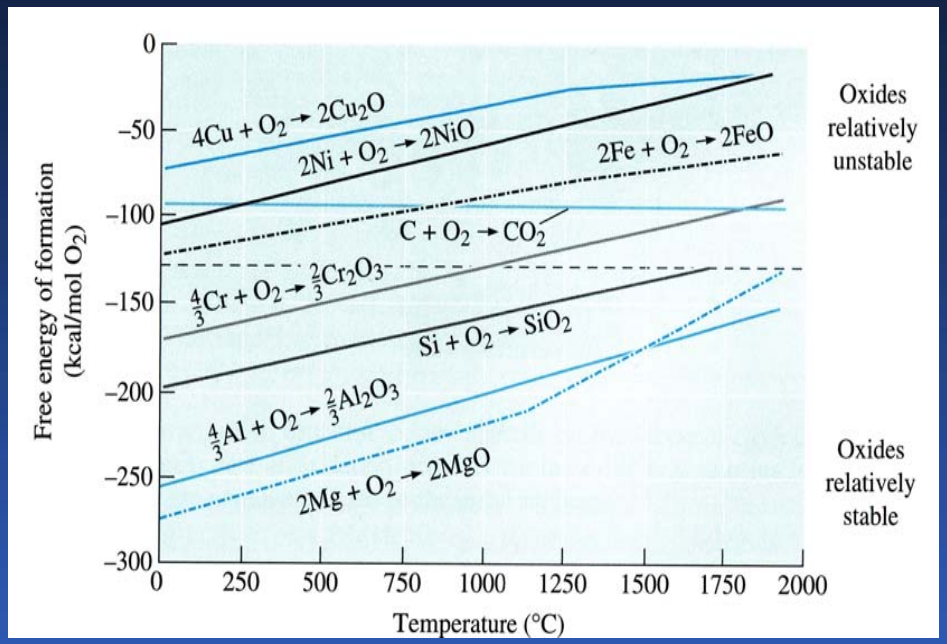


Nicholls et al. Surf. Coat. Technol. 149 (2002) 236–244
Z. Yu et al. Surf Coat. Technol. 201 (2006) 2326–2334

xhuang@mae.carleton.ca

Oxide Formers





Stable – large negative free energy of formation

Dense – no appreciable volume changes

Slow growth rate

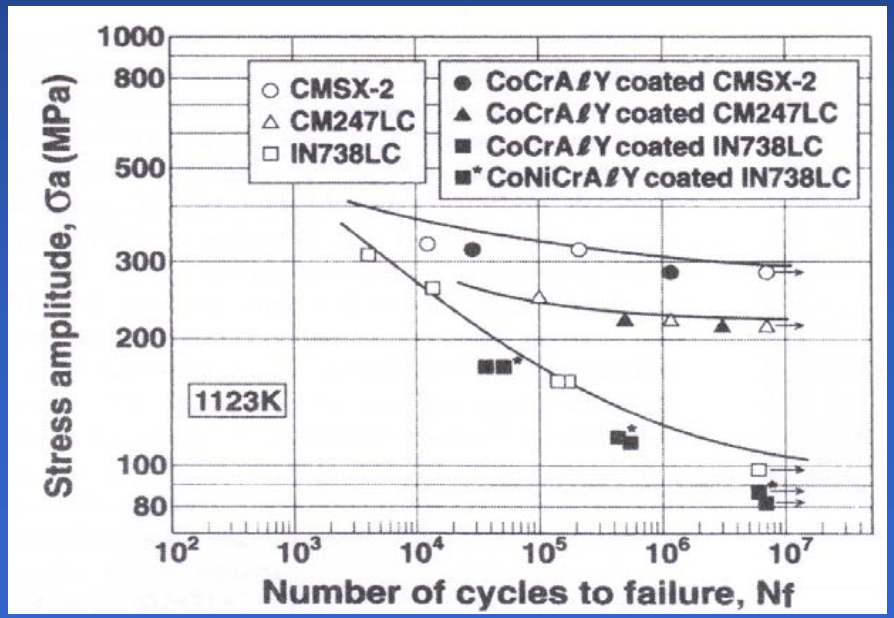
Carleton UNIVERSITY

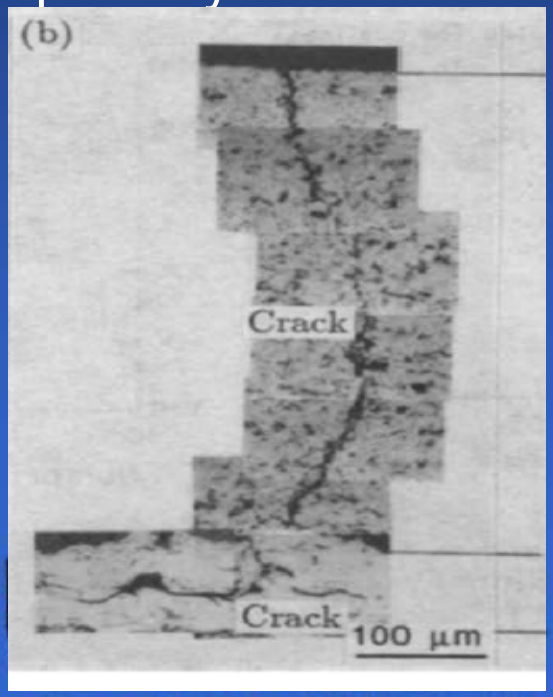
The science and engineering of materials, D. R. Askeland and P.P. Phule, 5th ed. Thomson

xhuang@mae.carleton.ca

Detrimental Effect of TBC to Substrates

- Fatigue life reduction
  - Crack initiates in the coating during cyclic loading
  - Propagation to substrate
- Microstructure degradation by internal stresses (coating process and operation)
  - γ rafting (anisotropic coalesence)
  - SX recrystallization
- Secondary reaction zone with single crystal superalloy





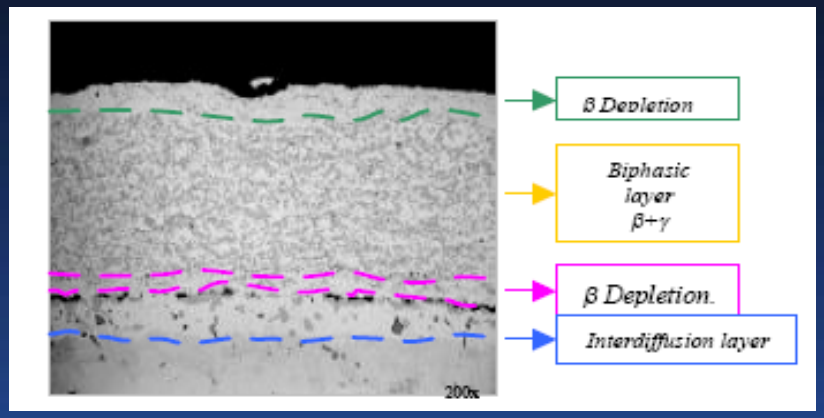
Suresh et al, Fatigue cracking in materials with brittle surface coatings 29(1993) 237-242

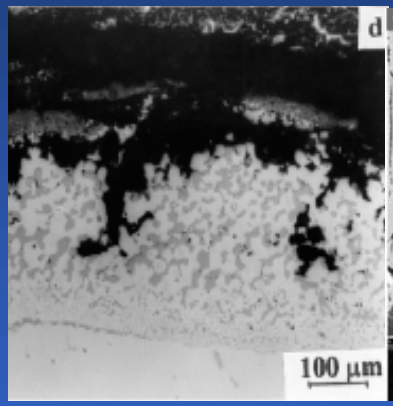
Itoh et al, Fatigue Fract Eng Mater Struct, 24(2001) 843-854

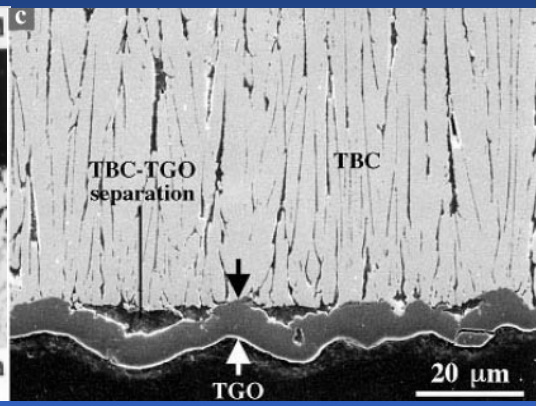
xhuang@mae.carleton.ca

Bond Coat Related Failures

- As standalone corrosion/oxidation resistant coating
  - Al forms Al_2O_3 (thermally grown oxide-TGO) during application, post coating heat treatment and during service
  - Al consumption due to TGO and inter-diffusion
  - When used alone TGO spalls and rebuilt until reaching critical Al content
  - Hot corrosion (molten salt)
- As bond coat
  - TGO thickens over time (Top coat provide oxygen sources due to defects such as pores, cracks and oxygen vacancies)
  - Mixed oxide formation
  - Internal strain buildup







Cracking
Spallation
Failure of coating system

Carleton UNIVERSITY

Turbo Expo 2006, GT2006-90748
Gurrappa, Surface and Coatings Technology 139 (2001) 272-283
Sohn et al. Surface and Coatings Technology 146 –147 (2001) 70–78

xhuang@mae.carleton.ca

Life Predication Approaches

- Existing Models:
  - TGO and critical crack size
  - Failure maps
  - COSIM (critical Al concentration)
  - Heckel (critical Al concentration)
  - DICTRA software – inter-diffusion microstructure map
- Materials, processes and service specific
- Semi-empirical
- Mostly based on TGO thickness and cumulative mechanical strain
- Calibrated by experimental data
- Lack of knowledge in damage mechanism, physical and mechanical properties of the material system as function of temperature and strain rate
- Lack of oxidation and diffusion models for multiple component system
- Material property changes must to incorporated into the lifing models

Goswami et al. High Temperature Materials and Processes, Vol.23(3) (2004)211-236
I.T. Spitsber et al. Materials Science and Engineering A, 394(2005) 176-191
A.G. Evans et al., Prog. Mater. Sci., **46** (2001) 505-553
R.S. Heckel et al. Thin Solid Film, 45(1977) 367
J.A. Nesbitt et al. Metall. Trans., 18A (1987) 2075
K.V. Dahl et al. Defect and Diffusion Forum Vols. 258-260 (2006) 73-78

Carleton UNIVERSITY

xhuang@mae.carleton.ca

Examples of Life Assessment

- For stand alone MCrAlY or aluminides – chemical failure
  - Aluminum consumption(8 at% Al=coating life)
  - % of beta phase (0% β=limit)
  - Diffusion layer thickness (model GT)

$$d = A\exp(\frac{-Q}{RT})t^n$$

- For TBC system – mechanical failure
  - Stress based (CTE mismatch, TGO growth stress, external…)

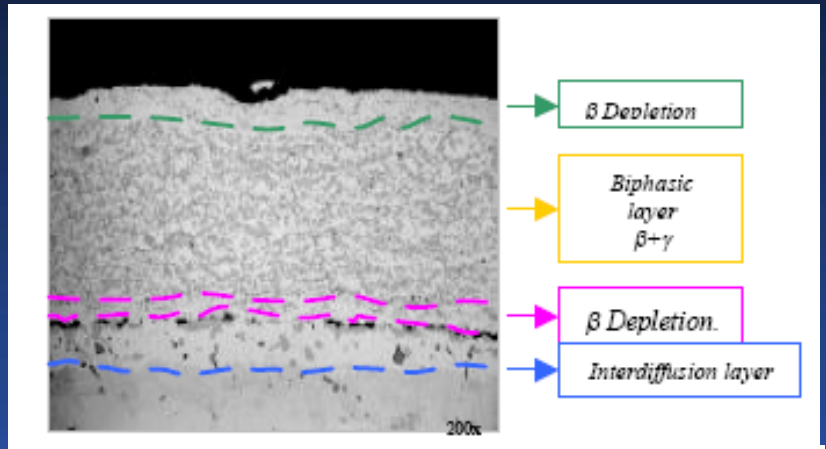
$$\sigma_{thermal} + \sigma_{external} + \sigma_{internal} > \sigma_{critical}$$

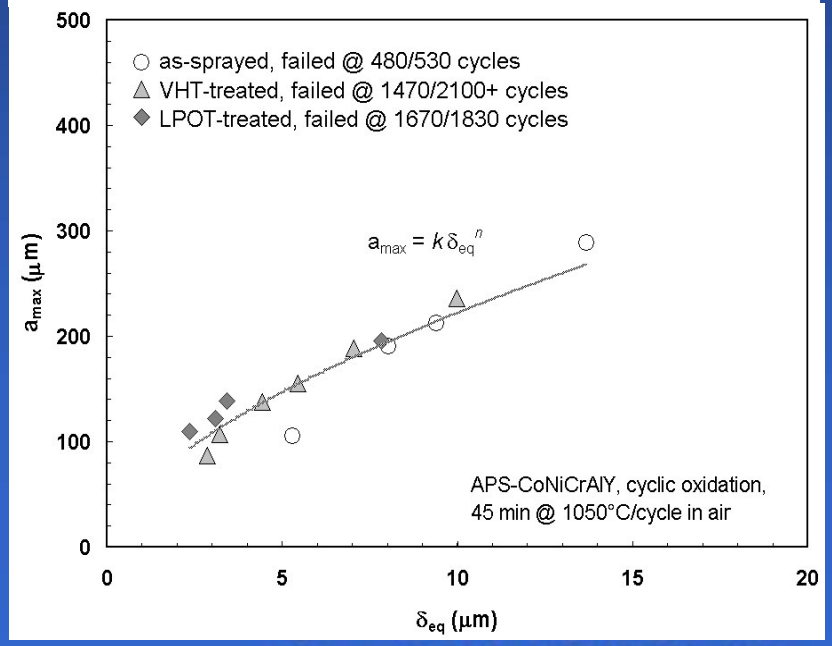
  - Strain based: cyclic loading, out-of-phase compressive strain determines the total fatigue life of TBC

$$N_f = k(\Delta\varepsilon_m^{tot})^n$$

$$\Delta\varepsilon_m^{tot} = \text{total mechanical strain}$$

- Empirical or experimental based method





P. Krukovsky, Surface and Coatings Technology 177 –178 (2004) 32-36
P. Kennard Wright, Materials Science and Engineering, A245(1998) 191-200
Turbo Expo 2006, GT2006-90138
Weijie Chen, et al, ITSC, 2009

xhuang@mae.carleton.ca

Challenges

- Material system - TBC less than ideal
  - TC: oxygen and radiation transparent, microstructure instability and limited thermal conductivity for EB-PVD coating
  - BC: higher CTE, brittle and high DBTT, use of precious metals (Pt, Rd…), composition variations
- Process
  - Lack of standard microstructure (process and vendor sensitive)
  - Complexity and variation of coating application techniques
- Life predication
  - Semi-empirical in nature
  - Each case must be validated by engine experience and field feedback
  - No universal model

Carleton UNIVERSITY

xhuang@mae.carleton.ca

Points for Discussion

- Development of new materials and processes?
- Reliable life predication method?
- In-situ monitoring?
- Self-healing coating system?

Carleton UNIVERSITY

Goswami et al. High Temperature Materials and Processes, 23(3) (2004)211-236

Acknowledgement

- Researchers at NRC: Drs. P. Patnaik, P. Au, R. Kearsey, and Weijie Chen
- Graduate students: Dongmei Wang, Jules Petitjean, Andrey Zakurdaev

Contact: xhuang@mae.carleton.ca
Tel: (613) 520 2600 ext. 5707

Carleton UNIVERSITY